

Highlight What You Want: Weakly-Supervised Instance-Level Controllable Infrared-Visible Image Fusion

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Abstract

Infrared and visible image fusion (VIS-IR) aims to integrate complementary information from both source images to produce a fused image with enriched details. However, most existing fusion models lack controllability, making it difficult to customize the fused output according to user preferences. To address this challenge, we propose a novel weakly-supervised, instance-level controllable fusion model that adaptively highlights user-specified instances based on input text. Our model consists of two stages: pseudo-label generation and fusion network training. In the first stage, guided by observed multimodal manifold priors, we leverage text and manifold similarity as joint supervisory signals to train text-to-image response network (TIRN) in a weakly-supervised manner, enabling it to identify referenced semantic-level objects from instance segmentation outputs. To align text and image features in TIRN, we propose a multimodal feature alignment module (MFA), using manifold similarity to guide attention weight assignment for precise correspondence between image patches and text embeddings. Moreover, we employ spatial positional relationships to accurately select the referenced instances from multiple semantic-level objects. In the second stage, the fusion network takes source images and text as input, using the generated pseudo-labels for supervision to apply distinct fusion strategies for target and non-target regions. Experimental results show that our model achieves state-of-the-art fusion performance and accurately highlights user-defined instances. Code: <https://github.com/GMY628/RIS-Fuse>.

1. Introduction

Infrared-visible image fusion (VIS-IR) combines complementary features from infrared (IR) and visible (VIS) images to create more informative results [39, 40, 53]. IR

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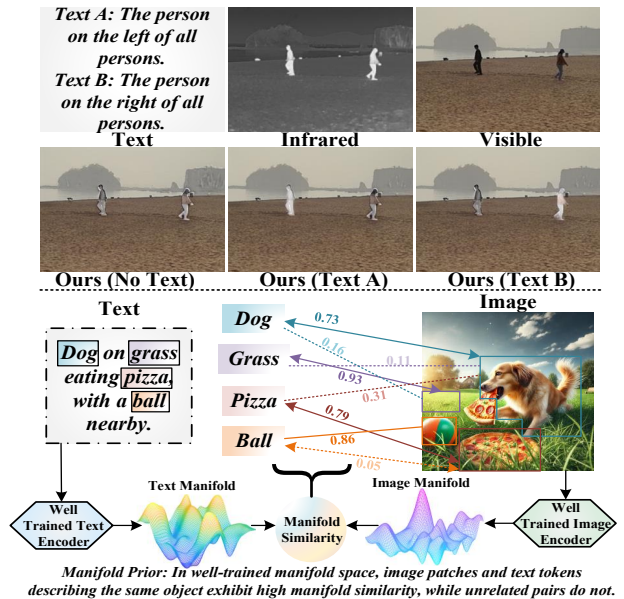


Figure 1. Top: Our model highlights instance objects guided by text. Bottom: Discovered manifold prior between text and image.

images capture thermal radiation, enabling effective low-light and night-time detection, but lack texture details, while VIS images offer rich details but are limited by lighting conditions. Fusing these modalities aims to preserve thermal information from IR while enhancing it with the detail from VIS [28, 58]. This is crucial for applications like surveillance, target detection, and autonomous driving [1, 26, 44, 52].

Existing fusion methods face major limitations in controllability and user adaptability. Users, real-world scenarios, or diverse downstream tasks, frequently require varying levels of fusion and target emphasis. However, once trained, these static methods produce fixed outputs, failing to adapt to dynamic needs [4]. While recent works have attempted to introduce controllability into fusion through text-driven

models [4, 19, 27, 42, 51, 54], these models are limited in scope, focusing on global adjustments or semantic-level highlighting, rather than enabling precise, instance-level control. Instance-level fusion control not only addresses dynamic user needs but also enhances downstream task performance, such as targeted object detection.

Achieving controllable image fusion (CIF) requires precise instance localization from text input. While existing image-text interaction paradigms enable instance localization in natural images [25, 30, 46], they are not directly applicable to VIS-IR images, as they are designed for single-modality natural images and overlook unique characteristics of infrared and visible modalities. For visible images, low-light conditions often obscure key objects, making localization unreliable. Infrared images, though effective in highlighting thermal information, lack texture, color, and fine-grained details [22], leading to two key challenges that hinder localization: (1) Non-thermal objects (e.g., background elements or low-temperature regions) lack distinctive features, making it harder to identify their categories. (2) Objects of the same category (e.g., people, vehicles) often share similar thermal signatures, lacking unique visual cues, which causes difficulty in instance differentiation. Despite the need, text-driven instance localization tailored for VIS-IR remains unexplored, limiting CIF advancements.

To address these challenges, we propose a weakly-supervised instance-level controllable fusion model that allows users to highlight the specified instance in fused images via text input, as shown in Fig. 1. Our model follows a two-stage training paradigm. Stage I employs weak supervision to generate pseudo-labels that capture instance localization from textual input, addressing the lack of instance-level annotations in existing VIS-IR datasets, without manual labeling; stage II trains the fusion network using these pseudo-labels, equipping it with both instance localization and image fusion capabilities. Stage I consists of a text-to-image response network (TIRN) and an instance selection module (ISM). TIRN identifies semantic-level objects based on user input and is trained in a weakly supervised manner using only manifold similarity (MS) and text as joint supervision signals [25, 62]. This is supported by an observed manifold prior: in well-trained feature spaces, for images and text represented as two separate manifolds [14, 36], image patches and text tokens describing the same object exhibit high MS, while unrelated pairs do not, as illustrated in Fig. 1. Moreover, inspired this prior, we tailor a multimodal feature interaction module (MFA) for VIS-IR-Text interactions in TIRN. MFA treats cross-modal misalignment as distortions in high-dimensional manifolds. By computing MS between image patches and text tokens, it leverages this similarity as a key factor for assigning attention weights, which significantly reduces the difficulty of instance object localization. After TIRN, ISM selects the

referenced instance based on spatial relationships. Spatial cues from text provide an effective basis for localization by comparing the coordinates of the “centroid” pixels of multiple objects, without requiring training.

In stage II, the image fusion network is trained with pseudo-labels, text, and source images, learning implicit associations between pseudo-labels and text while selectively highlighting referenced objects. To balance object emphasis and overall image quality, we design separate fusion strategies for target and non-target regions.

The contributions are summarized as follows.

- We present the first instance-level controllable VIS-IR fusion model that localizes user-specified objects from text and highlights target regions with tailored fusion modes.
- We propose TIRN and ISM, the first text-driven weakly supervised instance localization method for VIS-IR. TIRN generates semantic-level localization maps using only text and MS as supervision, while ISM refines them via spatial constraints to create pseudo-labels for fusion training.
- We design MFA that uses MS between image patches and text tokens as a crucial factor for matching features, promoting alignment from a manifold perspective.
- We discover a key manifold prior that justifies using MS to supervise TIRN and align text-image features.

2. Related Work

2.1. Uncontrollable vs. Controllable Image Fusion

Uncontrollable image fusion (UIF) methods operate in a static manner, applying uniform fusion strategies across the entire image, regardless of specific task requirements or user preferences. While effective for general feature merging, UIF lacks flexibility and may dilute critical details, reducing effectiveness for target detection. Most fusion models, such as [15, 17, 60, 61, 63], fall into this category.

Controllable image fusion (CIF) enables a dynamic fusion process, allowing users to specify their needs through textual descriptions and tailor fused images, offering advantages in tasks that require detailed object-level analysis. A few CIF models have been proposed. For example, Text-Fusion applies adaptive fusion to highlight user-specified semantic objects [4], demonstrating that even with known object locations, varying fusion levels improve downstream task accuracy. Similarly, TeRF enables region-level highlighting based on text [42]. However, these methods can only highlight semantic-level objects, not specific instances. Instance-level CIF remains a challenge.

Our Model. We propose a weakly-supervised instance-level CIF model that highlights user-specified instances.

2.2. Referring Image Segmentation

Referring Image Segmentation (RIS) focuses on segmenting specific objects within an image based on text descrip-

tions [24, 29, 34]. While RIS has advanced significantly for natural images [13], leveraging RGB cues like color, texture, and shape, it faces challenges with IR or VIS images.

(1) Modality Differences. IR images capture thermal radiation but lack RGB’s texture and color [20], making object distinction difficult. VIS images provide details but struggle in low-light conditions, where key targets may be obscured [10, 43]. (2) Reduced Instance Differentiation. RIS models excel with natural images by capturing semantic and instance-level details. However, in VIS-IR images, instance-level features are reduced, making it hard to distinguish multiple instances of the same class. For instance, a model may recognize “vehicles” but struggle to distinguish individual instances [4, 42], due to limited visual details.

To address these issues, we design a weakly-supervised instance localization method tailed for VIS-IR images.

3. Method

3.1. Overview

Fig. 2 shows our model with two stages: pseudo-label generation in a weakly-supervised manner and fusion network training. Stage I is the core of our approach, establishing a pixel-wise association between the image content and the input referring text without requiring pixel-level annotations. This stage includes the TIRN and the ISM. Once trained, the model, with frozen parameters, generates reliable masks serving as pseudo-labels for stage II. In stage II, the fusion network takes VIS-IR images and text as inputs, using the generated pseudo-labels as supervision and location references. By defining different learning objectives for target and non-target regions, the network learns to highlight specified instances based on user input.

Stages I and II are independently trained, each with a separate network and no shared modules or parameters. Stage I generates pseudo-labels for stage II, serving as training supervision. During inference (fusion), only the image fusion network is executed, as it has already learned both instance localization and fusion based on textual input.

3.2. Text-to-Image Response Network (TIRN)

As the first part of pseudo-label generation, TIRN’s goal is to assign high response values to all objects in multi-source images that match the semantics of the input text, while suppressing responses to irrelevant instances. TIRN enables the localization of semantic-level objects, providing a foundation for localizing instance-level objects in next step.

Inference. TIRN takes a pair of infrared image I_A , visible light image I_B , and user-provided text T as inputs. The inference workflow is illustrated in Fig. 2. First, I_A , I_B , and T are fed into image and text feature extractors, producing vision feature maps $F_V^A, F_V^B \in \mathbb{R}^{H \times W \times C_d}$ and text features $F_T \in \mathbb{R}^{1 \times C_d}$. Next, these features are processed

through the proposed MFA (detailed in Section 3.4), generating aligned features F_V' and F_T' . Note that F_V' integrates both F_V^A and F_V^B . Then, each pixel of F_V' is multiplied with each query from F_T' to produce a response map $R \in \mathbb{R}^{H \times W}$. The response between the i -th flattened pixel of F_V' and the j -th query of F_T' is computed as follows:

$$R_{i,j} = \sum_{z=1}^{C_d} F_V'(i, z) \cdot F_T'(j, z). \quad (1)$$

A higher response value in R indicates that the pixel is more likely to belong to the target semantic object, and vice versa. Finally, the average response S_x for each instance x in the instance segmentation map (denoted by S) of OMG-Seg [21] is calculated using Eq. (2):

$$S_x = \frac{\sum_{p \in P_x} R(p)}{|P_x|}, \quad (2)$$

where P_x represents the set of pixels forming instance x , and $|P_x|$ is the total number of pixels in instance x . Noting that most irrelevant instances yield small non-zero response values, we set a threshold 0.1. Any S_x values below this threshold are set to 0. The remaining instances with non-zero S_x values form the set of semantic-level objects P :

$$P = \{x | S_x \neq 0\}. \quad (3)$$

Thus, TIRN achieves localization of semantic-level objects.

Training. Inspired by [25], we adopt a classification-based approach, allowing the model to learn how to select the positive text from a set of positive and negative texts, optimizing the text-to-image response map to localize semantic-level objects. Specifically, we use text and text-image manifold similarity as joint supervision signals to perform this classification. For each pair of multi-source images I_A and I_B , we construct positive text and negative text samples. Positive texts describe the objects in I_A and I_B , while negative samples describe objects in other images. The input includes I_A , I_B , and a positive text Q^p , along with N randomly sampled negative texts Q^N . For each text query Q_j , we compute its image-level score g_j :

$$g_j = \max_i R_{i,j}^a + \frac{1}{HW} \sum_{i=1}^{HW} R_{i,j}^a + \psi(R_{:,j}^a), \quad (4)$$

where $\psi(R_{:,j}^a)$ is a regularization term [35]. A higher score indicates a better match between the text and the images. The model suppresses weights for negative texts and enhances those for positive texts by the loss function:

$$L_{TIRN} = L_{cls} + \lambda L_{ms}. \quad (5)$$

L_{cls} is the standard classification loss, described by:

$$L_{cls}(g, z) = \frac{1}{N+1} \sum_{j=1}^{1+N} z_j \log\left(\frac{1}{1-e^{-g_j}}\right) + (1-z_j) \log\left(\frac{e^{-g_j}}{1+e^{-g_j}}\right), \quad (6)$$

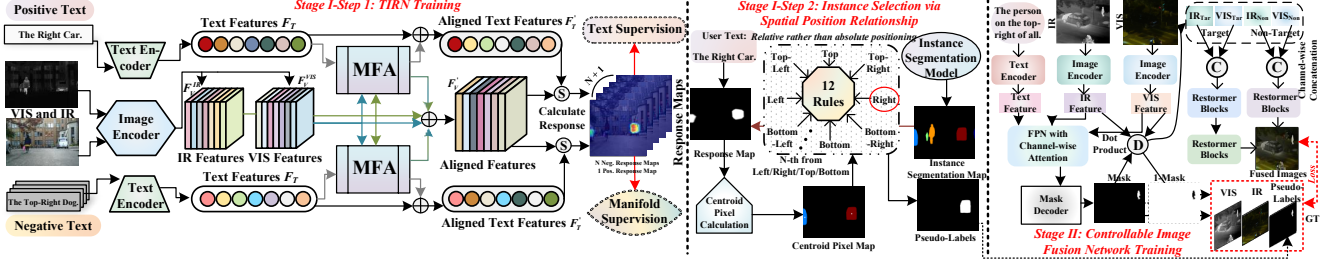


Figure 2. Schematic diagram of the proposed model. During training, our model operates in two stages: pseudo-label generation and fusion network training. The first stage consists of two main steps: text-to-image response (TIRN) and instance selection.

where $z \in \mathbb{R}^{1 \times (1+N)}$ represents supervision signals, with 1 indicating positive text and 0 indicating negative text. L_{ms} first computes the manifold similarity M between matched texts and images in a well-trained feature space. It then uses this similarity as a supervision signal to adjust the attention map A in MFA, supported by manifold priors, as follows:

$$L_{ms} = \alpha \sum_{i,j} \|A_{i,j} - f(M_{i,j})\|_2 - \beta \sum_{i,j} \|\text{Var}(A_{i,:}) - \text{Var}(M_{i,:})\|_2, \quad (7)$$

where f represents sigmoid; details on computing manifold similarity and distances are provided in Section 3.4. The first term ensures consistency between A and $f(M)$, guiding attention distribution according to manifold structure, while the second term prevents over-reliance on manifold similarity, ensuring diverse feature representation. As this loss decreases, text tokens with high manifold similarity to image patches receive greater attention, improving sensitivity to positive samples while distinguishing negative ones.

3.3. Instance Selection Module (ISM)

ISM selects the user-specified instance from the set P . Given that distinguishing between instance-level objects in infrared or visible images is challenging, we use the relative positional relationships provided in text as the basis for selection. First, the module computes each instance’s centroid pixel in P by averaging its pixel coordinates, as follows:

$$h_i = \left(\frac{1}{n} \sum_{j=1}^n x_j, \frac{1}{n} \sum_{j=1}^n y_j \right), \quad (8)$$

where n represents the total number of pixels in the i -th instance, and (x_j, y_j) denotes the pixel coordinates. This results in a set of centroid coordinates $H = \{h_1, h_2, \dots, h_k\}$, where k is the number of instances. Next, the model identifies position-related keywords, such as “left” or “bottom-right.” Finally, the module compares the centroid coordinates of each instance and selects the target instance h^* according to 12 designed rules, as described by Eq. (9).

$$h^* = \mathcal{R}_i(D_i), \quad (9)$$

where D_i denotes the detected spatial term in the vocabulary set D , and $\mathcal{R}_i$ represents the designed spatial relationship rule for D_i . D includes 12 position-related terms: left,

right, top, bottom, top-left, top-right, bottom-left, bottom-right, and the N -th from the left/right/top/bottom. The first eight words apply when the referenced object’s position is easily described, while the last four words address more complex cases, such as selecting an intermediate object from multiple instances. Each term corresponds to a strict, custom instance selection rule (details in appendix). By using relative positioning instead of absolute positioning, any instance can be accurately referenced.

This process shows that ISM selects the user-specified instance from multiple candidates without extra training, generating pseudo-labels for fusion network training.

3.4. Manifold-Based Feature Alignment Module

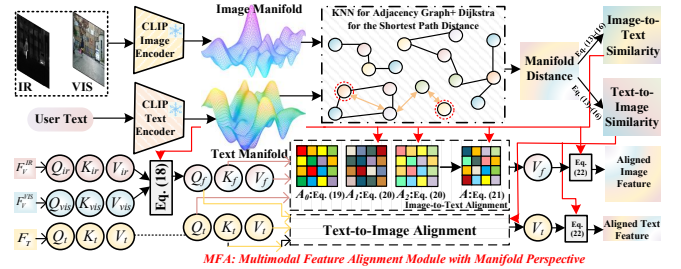


Figure 3. Schematic diagram of MFA. Manifold prior supports it.

To align multimodal feature in TIRN, we design MFA from a manifold perspective [14, 36, 47, 48]. MFA treats the three inputs—VIS image, IR image, and text—as separate manifolds. Supported by discovered manifold prior (see Fig. 1), the core design leverages manifold similarity as a guiding prior for alignment. MFA comprises three stages: computing manifold similarity, joint VIS-IR representation, and bidirectional alignment between text and image.

The first stage calculates the similarity matrices required for later steps. The second stage combines complementary features from VIS and IR images, avoiding mismatches between image objects and text caused by the limitations of a single source image, as discussed in Section 2.2. The final stages focus on aligning text and image features bidirectionally. The inputs include patch embeddings Q_{vis} , K_{vis} , V_{vis}

from F_V^{vis} ; Q_{ir} , K_{ir} , V_{ir} from F_V^{ir} ; and Q_t , K_t , V_t from text tokens, outputting aligned features F'_V and F'_T .

Manifold Similarity Computation. We compute manifold similarity matrices: M_{vis-ir} for VIS-IR alignment, and $M_{t \rightarrow v}$, $M_{v \rightarrow t}$ for text-image alignment. Due to space constraints, we take the computation of $M_{t \rightarrow v}$ as an example. We first extract features from I_A , I_B , and T using feature extractors, ensuring the effectiveness of manifold structure. Let the image features be $\mathbf{V} \in \mathbb{R}^{N \times d}$ and text features $\mathbf{T} \in \mathbb{R}^{M \times d}$, where N and M represent the number of patches and tokens, respectively. These features are mapped into a shared space and concatenated into $\mathbf{X}$:

$$\mathbf{X} = [\mathbf{V}; \mathbf{T}] \in \mathbb{R}^{(N+M) \times d}. \quad (10)$$

Next, we construct a KNN graph G from $\mathbf{X}$, capturing local relationships between features:

$$G = KNN(\mathbf{X}, k). \quad (11)$$

The manifold distance matrix $d_{t \rightarrow v} \in \mathbb{R}^{M \times N}$ is then computed, where $d_{t \rightarrow v}(j, i)$ represents the shortest path between text feature $\mathbf{T}_j$ and image feature $\mathbf{V}_i$ calculated by the Dijkstra algorithm [7] denoted by Γ :

$$d_{t \rightarrow v}(j, i) = \Gamma(G, N + j, i). \quad (12)$$

Then, we convert distances into similarity using RBF [2]:

$$M_{t \rightarrow v}^{fixed}(j, i) = \exp\left(-\frac{d_{t \rightarrow v}(j, i)^2}{2\sigma^2}\right), \quad (13)$$

where σ is the RBF scale parameter, controlling similarity. In order to prevent the fixed similarity matrix from hindering the training of the network, we perform an affine transformation on $M_{t \rightarrow v}^{fixed}$ to better fit the network:

$$M_{t \rightarrow v}^{learned} = W_0 \times M_{t \rightarrow v}^{fixed} + b, \quad (14)$$

where W_0 denotes the learnable matrix; b represents the learnable parameter. Finally, $M_{t \rightarrow v}^*$ is a linear combination of $M_{t \rightarrow v}^{fixed}$ and $M_{t \rightarrow v}^{learned}$ based on learnable parameter λ :

$$M_{t \rightarrow v}^* = \lambda \cdot M_{t \rightarrow v}^{fixed} + (1 - \lambda) \cdot M_{t \rightarrow v}^{learned}. \quad (15)$$

Finally, to align with subsequent computations, $M_{t \rightarrow v}^* \in \mathbb{R}^{M \times N}$ is reshaped through interaction with $\mathbf{T} \in \mathbb{R}^{M \times d}$, producing $M_{t \rightarrow v} \in \mathbb{R}^{N \times d}$, as follows:

$$M_{t \rightarrow v} = (M_{t \rightarrow v}^*)^T \times \mathbf{T}. \quad (16)$$

$M_{t \rightarrow v}$ guides the alignment from text to image. The computations of $M_{v \rightarrow t}$ and M_{vis-ir} follow a similar process.

Joint VIS-IR Representation. To obtain a comprehensive vision representation, we first globally pool M_{vis-ir} to derive M_{gap} . Then, linear projections generate parameters:

$$\alpha = \sigma(W_\alpha \times M_{gap} + b_\alpha), \quad \beta = \sigma(W_\beta \times M_{gap} + b_\beta), \quad (17)$$

where σ denotes the sigmoid function. Finally, the fused Q_f , K_f , and V_f are computed as:

$$Q_f = \alpha Q_{ir} + \beta Q_{vis}, K_f = \alpha K_{ir} + \beta K_{vis}, V_f = \alpha V_{ir} + \beta V_{vis}. \quad (18)$$

Text-to-Image Alignment. The process first computes the attention scores between $Q_t \in \mathbb{R}^{M \times d}$ and $K_f \in \mathbb{R}^{N \times d}$:

$$A_0 = \frac{Q_t \times K_f^T}{\sqrt{d_k}}. \quad (19)$$

Then, $M_{t \rightarrow v}$ interacts with Q_t and K_f , respectively:

$$A_1 = \frac{Q_t \times M_{t \rightarrow v}^T}{\sqrt{d_m}}, \quad A_2 = \frac{M_{t \rightarrow v} \times K_f^T}{\sqrt{d_m}}. \quad (20)$$

This step uses manifold similarity as a prior to guide attention toward potential text-image correspondences, enhancing feature alignment. These three attention maps are fused:

$$A = A_0 + \tau_1 \cdot A_1 + \tau_2 \cdot (W_1 \times A_2), \quad (21)$$

where τ_1 and τ_2 are learnable parameters that balance the importance of the three terms; $W_1 \in \mathbb{R}^{M \times N}$ denotes the learnable matrix. After applying Softmax to normalize A , the weights are used to compute the output:

$$F'_T = A \times V_f + \gamma \cdot (M_{t \rightarrow v} \times V_f). \quad (22)$$

The learnable γ ensures that $M_{t \rightarrow v}$ continues to impact output. This output serves as the aligned text features F'_T .

Image-to-Text Alignment. This process mirrors text-to-image alignment, with the following differences: (1) Q is derived from Q_f , while K and V are K_t and V_t ; (2) it uses the manifold similarity $M_{v \rightarrow t}$. The output serves as the aligned image features F'_V .

3.5. Image Fusion Network

In stage II, localization maps from stage I serve as pseudo-labels to train the fusion network. Training and inference are given here, with architecture presented in appendix.

Training. The network takes VIS, IR, and user-provided text as inputs, using pseudo-labels (binary masks) as location references. It employs two fusion modes to highlight target regions. For ROI in the pseudo-labels, effective features from both source images serve as supervision signals, including the brightness from IR and color and texture from VIS. To achieve this, we define the loss function with L_1 :

$$L_{ROI} = \varpi_1 L_1(Y_{FI}; IR) + \varpi_2 L_1(CB_{FI}; CB_{VIS}) + \varpi_3 L_1(CR_{FI}; CR_{VIS}) + \varpi_4 L_1(G_{FI}; G_{VIS}), \quad (23)$$

where Y_{FI} , CB_{FI} , and CR_{FI} are luminance and chrominance channels of fused images (via YCBCR decomposition); G denotes gradient calculation. 1st term enhances target instance brightness with IR, 2nd and 3rd terms ensure accurate color, and 4th term preserves textures, ensuring that highlights the object without compromising detail.

For non-target regions, we follow standard practices, using both source images as supervision signals:

$$L_{non-ROI} = \omega_1 L_1(FI; IR, VIS) + \omega_2 L_{ssim}(FI; IR, VIS). \quad (24)$$

In addition to L_1 , we use SSIM [45] in non-target regions.

We compute separate losses for ROI and non-ROI regions with pseudo-labels. To minimize loss, the network learns both fusion modes while localizing based on text.

Inference. The fusion network ζ_θ takes VIS images, IR images, and text as inputs and generates fused images FI :

$$FI = \zeta_\theta(IR, VIS, T). \quad (25)$$

4. Experimental Results

4.1. Setting

In stage 1, we set $\lambda = 500$, $\alpha = 1$, $\beta = 100$ and resize input images to 320×320 . We use AdamW (initial learning rate $5e^{-5}$) for 120 epochs, batch size 12. In stage 2, $\omega_1 = 50$, $\omega_2 = 4$, $\varpi_1 = 1$, $\varpi_2 = 2$, $\varpi_3 = 2$, $\varpi_4 = 5$. We use Adam with a learning rate of $1e^{-4}$, training for 100 epochs with batch size 2. Experiments are conducted on a machine with an RTX 4090 GPU, i9-14900K CPU, 128GB RAM, and PyTorch 2.3.0.

4.2. Dataset and Metrics

Both TIRN and the fusion network require training datasets. For TIRN, we use Li et al.’s dataset [4] with 2000 VIS-IR image pairs, each paired with three text descriptions referring to semantic-level objects. The fusion network’s training set includes source image pairs, instance-level text, and pseudo-labels generated in our model’s first stage, with re-structured text descriptions.

We select LLVIP [12], M³FD [26], MSRS [38], and TNO [41] as test sets, which are widely used in IF field. For evaluating controllable image fusion (CIF) models, each image pair is given a text description that randomly refers to an object in the image. All CIF models use the same image-text pairs for fair fusion results.

We use the metrics Q_{CB} [3], Q_Y [18], Q_E [33], Q_W [33], SF [45], AG [6], VIF [9] and $Q_{AB/F}$ [49]. SF and AG are unsupervised metrics; the rest are supervised. For supervised metrics, we follow convention by using each source image as ground truth for metric calculation.

4.3. Comparison with SOTA

Comparison Models. We select ten methods, including FusionGAN [31], DIDFuse [55], GANMcC [32], DeFusion [23], ReCoNet [11], CDDFuse [56], DDFM [57], LRRNet [16], FILM [59], and TextFusion [4]. The first nine models are uncontrollable fusion (UIF) methods, using only source images as input; while TextFusion and our model are controllable (CIF), using both source images and text inputs.

Qualitative Evaluation. Fig. 4 illustrates the qualitative comparison across three samples from different test sets. Both our model and TextFusion adjust the fused image

based on text. Our model uniquely highlights instance-level objects, whereas TextFusion emphasizes all objects matching the text. For example, in the first group, TextFusion fails to accurately isolate the instances described by “the car on the right” and “the person on the top left,” while our model precisely localizes the targets. “No text” means no text input, where our model applies only the non-target fusion mode, achieving promising results. This demonstrates its strong performance in static fusion.

Quantitative Evaluation. Table 1 reports the results of various models across eight metrics. Our model achieves superior results in most metrics, showing its effectiveness in instance localization and fusion quality.

4.4. Ablation Study

As our main innovation focuses on stage I, we measure response maps or pseudo-labels using segmentation metrics, with segmentation maps of referred instances from large models serving as GT. Quantitative results are presented, with qualitative results in appendix.

Effect of TIRN. TIRN is trained with text and manifold similarity supervision. To assess its impact, we conduct the following ablations: (1) removing text supervision; (2) removing the manifold similarity; (3) removing TIRN entirely; (4) the default setup. Table 2(I) shows that both supervision signals significantly improve model performance. TIRN is crucial for generating accurate pseudo-labels, as its removal results in a marked drop in pseudo-label accuracy.

Effect of MFA. To evaluate MFA, we perform the following ablations: (1) removing the module; (2) replacing it with cross-attention; (3) using random matrices instead of the manifold similarity matrix; (4) using well-trained CLIP instead of MFA; (5) the default setup. Results in Table 2(II) confirm that MFA, particularly the manifold similarity, is vital for enhancing model performance.

Effect of Instance Selection via Spatial Relationships. This module refines instance localization using positional descriptions. We perform: (1) removing the module; (2) reducing spatial rules from 12 to 4; (3) the default setup. Table 2(III) shows that without this module, pseudo-labels fail to capture instance-level objects, and fewer rules reduce localization accuracy for positions like “top-left”.

4.5. Validation of the Necessity of Tailored Instance Localization Method for VIS-IR

As discussed in Sections 1 and 2.2, RIS models designed for natural images are unsuitable for object localization in VIS-IR. To verify, we test several RIS models (LAVT [50], VLT [8], MG [5]) on IR and VIS images. Unlike these models, which process only single images and yield suboptimal results due to modality limitations, our VIS-IR-tailored method takes two images as input to create a joint visual representation, reducing localization difficulty. As shown

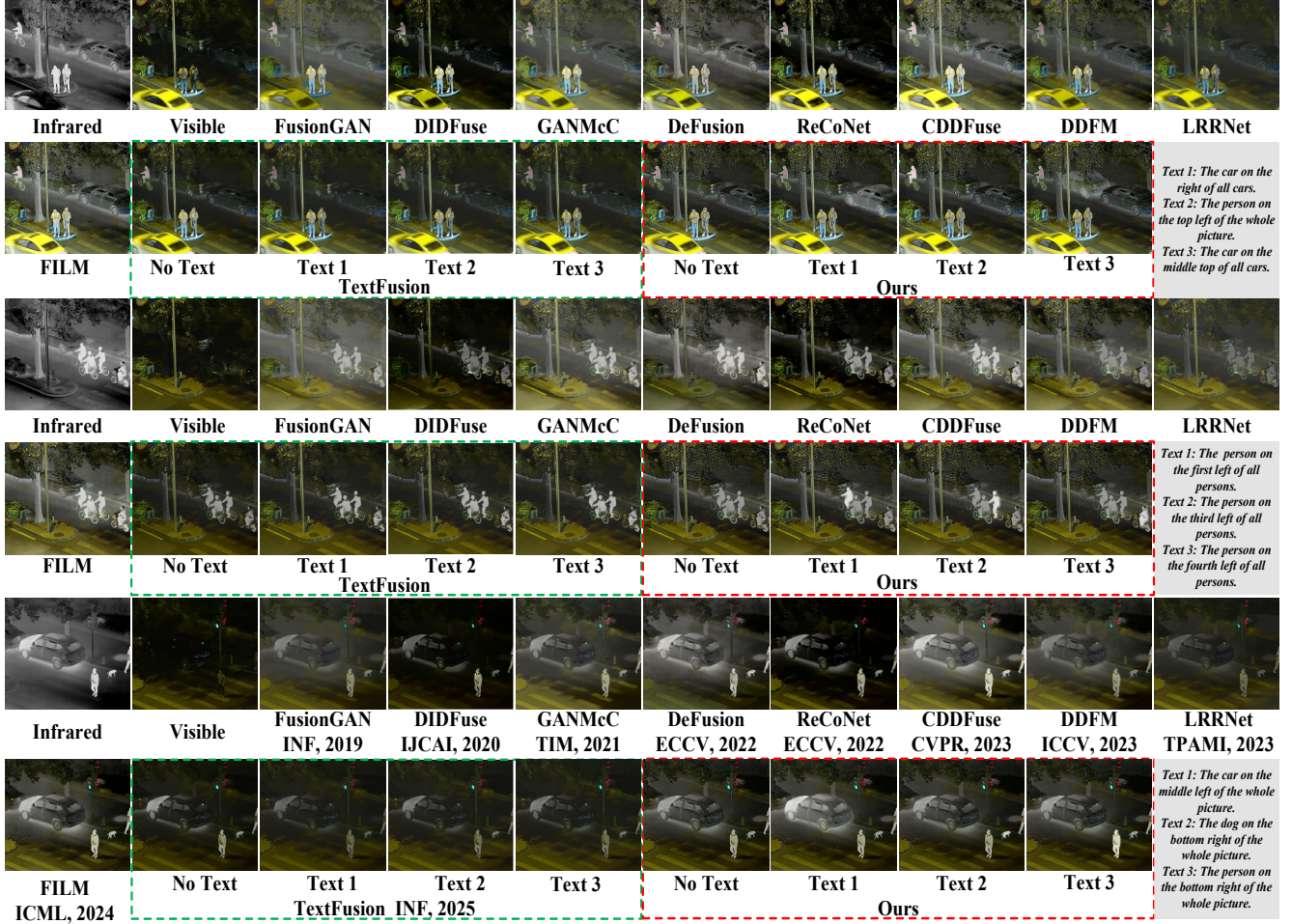


Figure 4. Qualitative comparison of fusion models. The first nine UIF models yield static results regardless of text. TextFusion and ours take one text and two source images as input. TextFusion highlights semantic-level objects, while our model targets the referenced instance.

Table 1. Quantitative comparison of various fusion models across four test sets.

Test Sets	Methods	$Q_{CB}\uparrow$	$Q_Y\uparrow$	$Q_E\uparrow$	$Q_W\uparrow$	$SF\uparrow$	$AG\uparrow$	$VIF\uparrow$	$Q_{AB/F}\uparrow$	Test Sets	Methods	$Q_{CB}\uparrow$	$Q_Y\uparrow$	$Q_E\uparrow$	$Q_W\uparrow$	$SF\uparrow$	$AG\uparrow$	$VIF\uparrow$	$Q_{AB/F}\uparrow$
LLVIP	FusionGAN	0.251	0.487	0.306	0.289	5.970	1.609	0.498	0.212	M ³ FD	FusionGAN	0.330	0.552	0.376	0.333	8.012	2.688	0.388	0.264
	DIDFuse	0.424	0.338	0.432	0.389	10.313	2.098	0.444	0.267		DIDFuse	0.445	0.653	0.668	0.654	15.423	5.306	0.669	0.511
	GANMcC	0.326	0.572	0.368	0.413	6.286	1.874	0.654	0.292		GANMcC	0.392	0.615	0.409	0.431	7.590	2.695	0.527	0.289
	DeFusion	0.432	0.734	0.576	0.650	8.940	2.523	0.776	0.466		DeFusion	0.427	0.672	0.502	0.467	8.408	2.945	0.549	0.345
	ReCoNet	0.407	0.395	0.611	0.551	9.915	2.774	0.580	0.408		ReCoNet	0.455	0.740	0.706	0.670	11.955	4.479	0.604	0.508
	CDDFuse	0.459	0.857	0.767	0.762	13.278	3.432	0.958	0.641		CDDFuse	0.475	0.870	0.793	0.722	16.491	5.417	0.781	0.632
	DDFM	0.406	0.613	0.370	0.451	5.785	1.869	0.714	0.300		DDFM	0.405	0.656	0.555	0.532	9.725	3.379	0.606	0.449
	LRRNet	0.360	0.521	0.512	0.411	8.874	2.277	0.558	0.405		LRRNet	0.435	0.721	0.646	0.574	26.690	9.269	0.762	0.461
	FILM	0.485	0.759	0.825	0.791	14.361	3.926	0.976	0.675		FILM	0.493	0.839	0.816	0.748	16.757	5.545	0.806	0.653
	TextFusion	0.455	0.692	0.654	0.552	11.765	2.894	0.738	0.495		TextFusion	0.477	0.713	0.755	0.662	16.836	5.523	0.619	0.544
MSRS	Ours	0.511	0.784	0.843	0.803	14.572	4.005	0.996	0.711	TNO	Ours	0.517	0.905	0.833	0.754	16.627	5.609	0.838	0.694
	FusionGAN	0.322	0.391	0.206	0.206	4.354	1.446	0.442	0.140		FusionGAN	0.408	0.539	0.386	0.373	6.269	2.362	0.418	0.224
	DIDFuse	0.407	0.238	0.432	0.402	9.644	2.006	0.304	0.202		DIDFuse	0.463	0.655	0.602	0.614	11.768	4.249	0.593	0.403
	GANMcC	0.424	0.574	0.404	0.444	5.664	1.999	0.635	0.302		GANMcC	0.437	0.607	0.420	0.445	6.217	2.513	0.513	0.275
	DeFusion	0.514	0.749	0.730	0.753	8.146	2.644	0.730	0.507		DeFusion	0.482	0.707	0.592	0.568	6.598	2.675	0.553	0.359
	ReCoNet	0.378	0.347	0.720	0.692	9.975	2.990	0.490	0.404		ReCoNet	0.463	0.673	0.592	0.610	7.958	3.353	0.531	0.373
	CDDFuse	0.567	0.827	0.868	0.859	11.556	3.734	1.051	0.693		CDDFuse	0.450	0.787	0.697	0.659	11.621	4.330	0.730	0.496
	DDFM	0.482	0.662	0.582	0.579	7.388	2.513	0.743	0.474		DDFM	0.437	0.485	0.473	0.484	8.128	3.213	0.371	0.292
	LRRNet	0.393	0.507	0.659	0.632	8.473	2.641	0.541	0.454		LRRNet	0.483	0.701	0.513	0.513	9.438	3.627	0.538	0.367
	FILM	0.569	0.822	0.873	0.863	11.726	3.858	1.056	0.723		FILM	0.483	0.821	0.745	0.726	12.579	4.556	0.725	0.529
	TextFusion	0.490	0.618	0.795	0.805	10.230	3.045	0.694	0.475		TextFusion	0.525	0.750	0.581	0.582	10.217	3.986	0.598	0.392
	Ours	0.570	0.875	0.887	0.876	11.788	3.877	0.997	0.698		Ours	0.520	0.829	0.746	0.733	12.935	4.791	0.704	0.566

in Fig. 5, these models fail to localize instances correctly, while our method (Stage I) produces accurate results. Numerical results are provided in supplementary material.

To show the impact of MFA, the right part of Fig. 5 visualizes TIRN outputs using MFA versus cross-attention, with MFA yielding higher response values for correct ob-

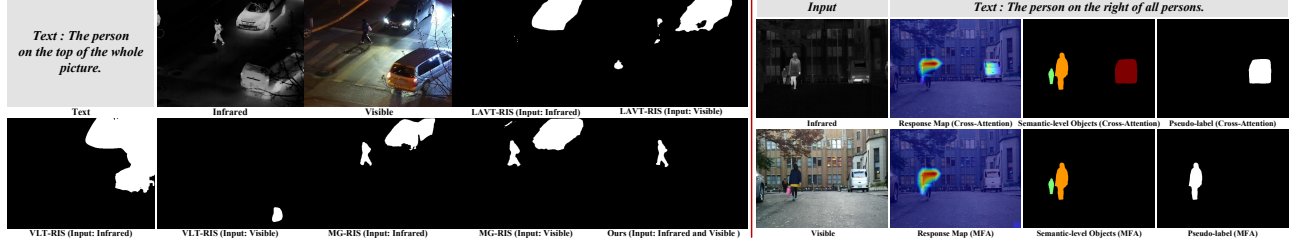


Figure 5. Visualization. The left part shows a comparison of our localization method with recent RIS models on VIS and IR images. The right part illustrates the impact of our MFA and cross-attention on pseudo-label accuracy.

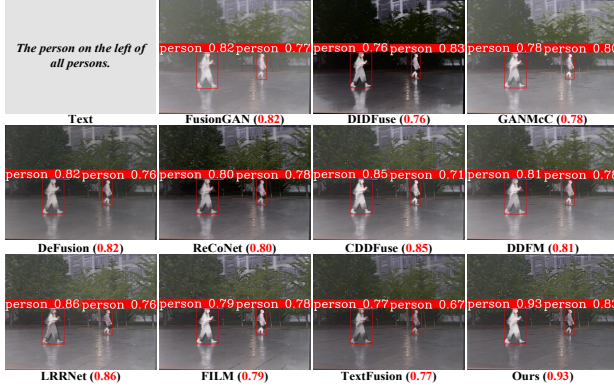


Figure 6. Qualitative comparisons of different fusion models on the TOD task. Fused images generated by our model enable YOLO-v5 to achieve higher confidence scores for target instances.

Table 2. Quantitative results of multiple ablation experiments. The checkmark indicates which of RM or PL is used to calculate metrics. MS: Manifold Similarity; CA: Cross Attention; RM: Response Map; PL: Pseudo-Label; SR: Spatial Relationships. PointIt and PointM can only be used for the evaluation of RM.

Descriptions	RM	PL	Metrics			
			P@0.5[37]	mIoU	PointIt[25]	PointM[25]
I w/o Text-Supervision	✓		0.240	1.231	9.886	6.027
w/o MS	✓		5.941	22.257	59.073	44.274
Default Setup	✓		23.437	41.714	83.394	75.526
II w/o MFA	✓		4.985	21.052	57.959	42.252
MFA→CA	✓		9.526	25.615	65.606	50.311
MS→Random Matrix	✓		2.258	11.790	21.974	13.141
MFA→CLIP	✓		4.661	20.713	56.816	41.227
Default Setup	✓		23.437	41.714	83.394	75.526
III w/o SR	✓		23.437	41.714	83.394	75.526
w/o TIRN		✓	32.283	30.212	-	-
Rules Reduce: 8→4		✓	60.629	54.579	-	-
Default Setup	✓		73.228	66.065	-	-

jects. It also shows the two-step pseudo-label generation: TIRN captures all semantic-level objects, and then instance selection module pinpoints the correct instance.

4.6. Downstream Task: Targeted Object Detection

Our model enhances TOD by highlighting referenced instances. To assess the impact of different fusion models on TOD, we use YOLO-v5 to detect objects in various fused images (see Fig. 6, Table 3). By emphasizing low-

Table 3. Comparison results of fusion models on the TOD task.

Fusion Methods	Recall	mAP		
		@0.50	@0.75	@[0.5;0.95]
FusionGAN [31]	0.207	0.307	0.179	0.170
DIDFuse [55]	0.272	0.402	0.250	0.231
GANMcC [32]	0.267	0.393	0.235	0.222
DeFusion [23]	0.259	0.378	0.233	0.214
ReCoNet [11]	0.287	0.423	0.252	0.243
CDDFuse [56]	0.292	0.430	0.261	0.246
DDFM [57]	0.281	0.422	0.245	0.234
LRRNet [16]	0.304	0.447	0.260	0.252
FILM [59]	0.300	0.45	0.261	0.251
TextFusion [4]	0.288	0.423	0.253	0.243
Ours	0.314	0.472	0.268	0.263

brightness instances, our model improves YOLO-v5’s confidence scores and achieves superior quantitative results on the M³FD dataset. Compared to other fusion models, its instance-level controllability offers distinct advantages in TOD, further underscoring its value.

5. Conclusion

We propose a novel instance-level controllable image fusion model for VIS-IR fusion, enabling users to highlight any instance via text. Lacking suitable datasets and labels, we employ a two-stage weakly supervised approach to generate pseudo-labels. In stage I, we train TIRN with text and manifold similarity as joint supervision, using MFA to align image patches with text tokens and applying twelve spatial rules for instance selection. In stage II, the fusion network locates target objects based on text input and applies two fusion modes. Results show accurate instance highlighting and state-of-the-art fusion performance by our model.

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Highlight What You Want: Weakly-Supervised Instance-Level Controllable Infrared-Visible Image Fusion

Supplementary Material

1. Spatial Position Rules

Assuming the coordinates of the centroid pixel h_i are (x_i, y_i) , we compare these coordinates based on positional descriptions to select the target instance that satisfies the given spatial relationship. We design 12 rules to address possible directional descriptions in the text. The first eight apply when the referenced object’s position is easily describable, while the last four handle more complex cases, such as selecting an intermediate object among multiple instances:

1. Up: Select the instance with the highest position, i.e., the one with the largest y_i :

$$i^* = \arg \max_i \{y_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

2. Down: Select the instance with the lowest position, i.e., the one with the smallest y_i :

$$i^* = \arg \min_i \{y_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

3. Left: Select the instance that is furthest to the left, i.e., the one with the smallest x_i :

$$i^* = \arg \min_i \{x_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

4. Right: Select the instance that is furthest to the right, i.e., the one with the largest x_i :

$$i^* = \arg \max_i \{x_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

5. Top-left: Select the instance that is furthest up and to the left, i.e., the one with the smallest x_i and the largest y_i :

$$i^* = \arg \max_i \{y_i - x_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

6. Top-right: Select the instance that is furthest up and to the right, i.e., the one with the largest sum of x_i and y_i :

$$i^* = \arg \max_i \{x_i + y_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

7. Bottom-left: Select the instance that is furthest down and to the left, i.e., the one with the smallest sum of x_i and y_i :

$$i^* = \arg \min_i \{x_i + y_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

8. Bottom-right: Select the instance that is furthest down and to the right, i.e., the one with the largest x_i and the smallest y_i :

$$i^* = \arg \min_i \{y_i - x_i\}, \quad \forall i \in \{1, 2, \dots, k\}.$$

9. N -th from the left: Select the N th instance from the left by sorting x_i in ascending order:

$$i^* = \arg \min_{i \in S_N} \{x_i\}, \quad S_N = \sigma_{\uparrow}(x_i), \quad \forall i \in \{1, 2, \dots, k\},$$

where $\sigma_{\uparrow}$ denotes the ascending function.

10. N -th from the right: Select the N th instance from the right by sorting x_i in descending order:

$$i^* = \arg \max_{i \in S_N} \{x_i\}, \quad S_N = \sigma_{\downarrow}(x_i), \quad \forall i \in \{1, 2, \dots, k\},$$

where $\sigma_{\downarrow}$ denotes the descending function.

11. N -th from the bottom: Select the N th instance from the bottom by sorting y_i in ascending order:

$$i^* = \arg \min_{i \in S_N} \{y_i\}, \quad S_N = \sigma_{\uparrow}(y_i), \quad \forall i \in \{1, 2, \dots, k\}.$$

12. N -th from the top: Select the N th instance from the top by sorting y_i in descending order:

$$i^* = \arg \max_{i \in S_N} \{y_i\}, \quad S_N = \sigma_{\downarrow}(y_i), \quad \forall i \in \{1, 2, \dots, k\}.$$

Finally, it is necessary to emphasize that our spatial relationships represent the relative rather than absolute positioning between multiple objects. We do not rigidly divide the image into fixed regions. For example, two people in the right area can be referred to as the “left” or “right person.”

2. Architecture of Image Fusion Network

Holistically, our image fusion network receives multi-source images and text as input, and uses pseudo labels as supervisory signals. Specifically, this network consists of two stages: outputting the mask based on text localization instances and implementing two different fusion modes according to the mask.

First, we employ Restormer [10] blocks to extract features of the infrared image I_{ir} and the visible image I_{vis} , as follows:

$$F_{ir} = E_{ir}(I_{ir}), \quad F_{vis} = E_{vis}(I_{vis}), \quad (1)$$

where E_{ir} and E_{vis} denote the infrared visual feature encoder and the visible light visual feature encoder, respectively. Next, the extracted visual features F_{ir} and F_{vis} are concatenated and passed through the visual feature fusion module $\mathcal{U}$ to obtain a unified visual feature V .

$$V = \mathcal{U}(F_{ir} \oplus F_{vis}), \quad (2)$$

where $\mathcal{U}$ represents the interaction operation via Restormer blocks [10]. Simultaneously, we utilize CLIP’s text encoder with frozen weights to process the text t and obtain the text feature F_t , as follows:

$$F_t = CLIP_{text}(t), \quad (3)$$

where $CLIP_{text}$ denotes CLIP’s text encoder.

Then, we construct an FPN (Feature Pyramid Network) using convolutional layers to enable multi-scale interaction between the text feature F_t and the unified visual feature V , ultimately yielding the positioning map P , i.e.,

$$P = \mathcal{S}(\mathcal{F}(V, F_t)), \quad (4)$$

where $\mathcal{F}$ represents the FPN, $\mathcal{S}$ represents the sigmoid operation. P is a binary image representing the location of the object referred to by the text. The white region indicates the ROI (Region of Interest), while the black region represents the non-ROI.

Subsequently, we utilize the positioning map P to perform Hadamard product with F_{ir} and F_{vis} , obtaining the feature representations of the ROI and non-ROI regions from the two source images, denoted by F_{ir_ROI} , F_{vis_ROI} , $F_{ir_non-ROI}$, and $F_{vis_non-ROI}$. Since different fusion modes are applied to the ROI and non-ROI regions, this step prepares for their independent fusion, preventing interference between the two regions.

$$F_{ir_ROI} = P \odot F_{ir}, \quad (5)$$

$$F_{vis_ROI} = P \odot F_{vis}, \quad (6)$$

$$F_{ir_non-ROI} = (1 - P) \odot F_{ir}, \quad (7)$$

$$F_{vis_non-ROI} = (1 - P) \odot F_{vis}, \quad (8)$$

where $\odot$ denotes the Hadamard product.

Next, we use Restormer blocks M_{ROI} to fuse the features of ROI regions, as follows:

$$F_{ROI} = M_{ROI}(F_{ir_ROI}, F_{vis_ROI}). \quad (9)$$

For non-ROI regions, we also use Restormer blocks $M_{non-ROI}$ but without shared weights, as follows:

$$F_{non-ROI} = M_{non-ROI}(F_{ir_non-ROI}, F_{vis_non-ROI}). \quad (10)$$

Finally, we input F_{ROI} and $F_{non-ROI}$ into the image reconstruction decoder D which consists of Restormer blocks to obtain the final fused image F :

$$F = D(F_{ROI}, F_{non-ROI}). \quad (11)$$

Clearly, we avoid directly computing the loss between the pseudo-labels and P ; instead, we use them as a positional reference. Through supervision from two loss functions during training (Eq. X-Eq. X in the main text), the network is forced to learn: (1) accurate localization based on the text, i.e., generating P that closely matches the pseudo-labels, and (2) correctly applying the two fusion modes. Therefore, during inference (fusion), our model only requires the image fusion network, without needing the stage I model.

3. Why is our multimodal feature alignment module necessary?

Although there are many established multimodal feature alignment methods, they are not fully applicable to our task due to the following reasons:

Task-Specific Requirements: Existing methods, such as those based on cross-modal attention mechanisms [6], canonical correlation analysis [3], and contrastive learning [7], typically aim to align global features or focus on aligning image and text representations in a general sense. However, our task requires fine-grained, instance-level alignment where specific object instances need to be localized and highlighted based on user input. Most existing methods fall short of addressing this level of specificity.

Manifold Structure of Multimodal Data: Previous methods generally assume a linear or simple metric space for alignment, but multimodal data, particularly in the case of image and text, often exist in complex, nonlinear manifold spaces [4, 8]. These methods fail to fully exploit the manifold structure that underlies multimodal interactions. In contrast, our approach leverages manifold similarity as a guiding prior, which captures the intrinsic geometric relationships between image patches and text tokens. This prior allows for more precise alignment, especially for tasks requiring high semantic accuracy.

Weakly Supervised Training: Many alignment approaches rely on large, fully annotated datasets with paired samples for supervised training. However, in our domain, instance-level annotations for multimodal datasets, such as VIS-IR image pairs with corresponding object labels, are either unavailable or highly limited. To address this, we propose a weakly-supervised alignment strategy that generates pseudo-labels using text-to-image similarity, avoiding the need for ground truth annotations.

Bidirectional Alignment: Existing methods predominantly align features in one direction (e.g., from text to image or vice versa). This unidirectional alignment is often insufficient when dealing with complex multimodal interactions where both modalities need to inform each other. Our method introduces bidirectional alignment, ensuring that both text and image features are jointly refined, enhancing the quality of the fusion output.

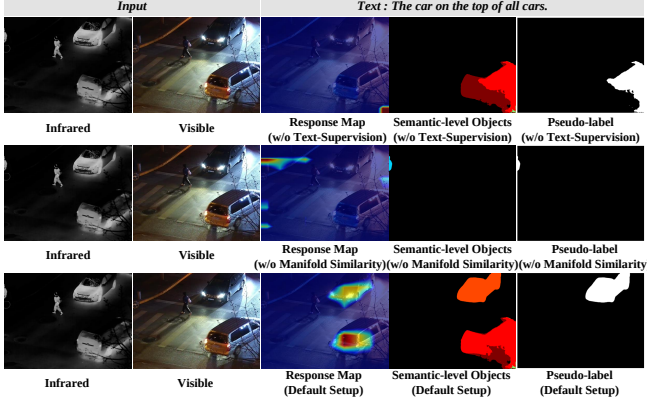


Figure 1. Qualitative results of ablation study on TIRN.

Thus, the unique combination of task-specific instance-level alignment, the utilization of manifold similarity as a geometric prior, weakly-supervised learning, and bidirectional feature alignment distinguishes our method from existing solutions, enabling more effective feature alignment.

4. Comparison with SOTA

Because of space constraints in the main paper, we provide additional qualitative comparison results for various VIS-IR fusion models here, as shown in Fig. 4. It is evident that the uncontrollable image fusion model cannot tailor fused images based on textual input, aligning with the analysis in the Introduction. For the controllable image fusion models, both TextFusion and our model are capable of generating images based on text. However, TextFusion can only highlight multiple objects at the semantic level without precise instance-level localization, whereas our model achieves this level of specificity.

5. Ablation Study

To effectively evaluate the first stage of our model, we construct a VIS-IR dataset that includes text descriptions referring to instance-level objects and binary masks (ground truth, GT) indicating their locations. To obtain these GT masks, we first utilize a large vision foundation model for segmentation [5] to generate preliminary instance segmentation maps. We then select the corresponding instances based on different textual cues and manually refine these maps to ensure accuracy. The numerical results of the ablation experiment, as listed in Table 2 of the main paper, are calculated based on these refined GT maps.

Here, we present qualitative comparison results of ablation experiments in the main paper. As shown in Fig. 1, Fig. 2, and Fig. 3, removing the proposed modules leads to a significant drop in the quality of response maps and pseudo-labels, demonstrating the effectiveness of TIRN, MFA, and

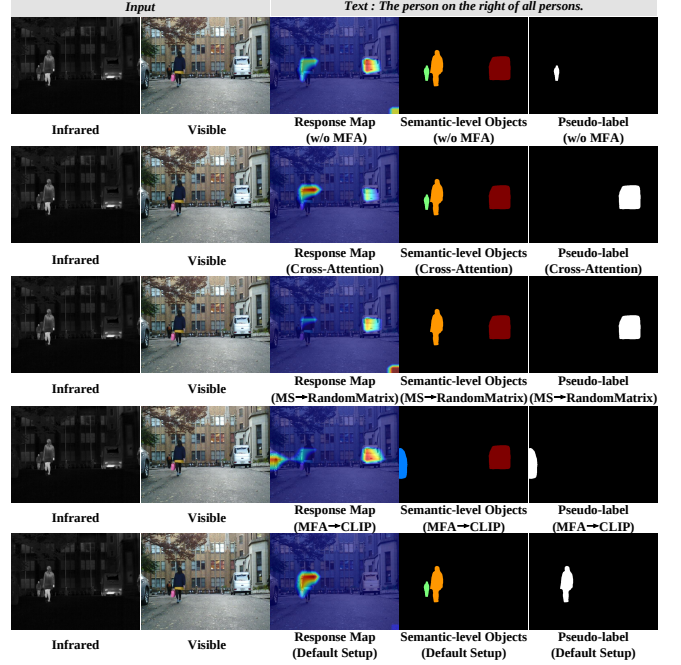


Figure 2. Qualitative results of ablation study on MFA.

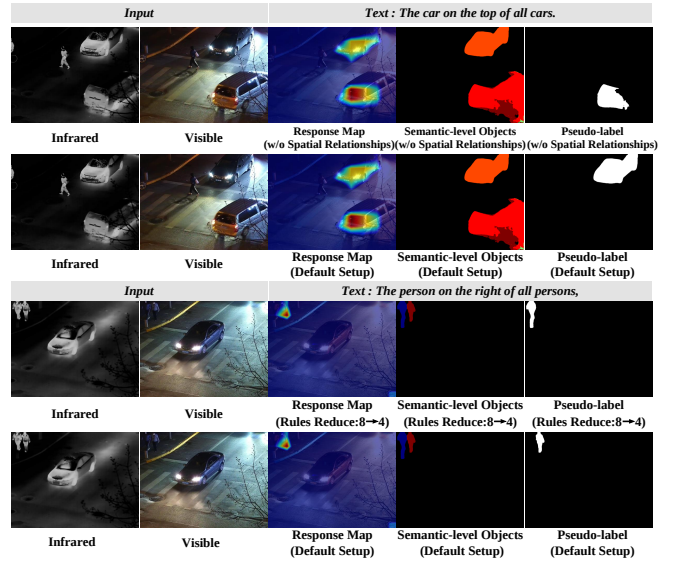


Figure 3. Qualitative results of ablation study on instance selection module.

the instance selection module.

6. Validation of the Necessity of Tailored Instance Localization for VIS-IR

As discussed in Sections 1 and 2.2 of the main paper, RIS models designed for natural images are unsuitable for object localization in VIS-IR. To verify, we test several RIS mod-

Table 1. Qualitative comparison of our localization method with RIS models on VIS and IR images. Existing RIS models, designed for natural images, can only take either infrared or visible images as single inputs, lacking the ability to integrate complementary information from both modalities for localizing referred objects.

Methods	Precision@0.5	MIoU
LAVT-RIS(Input:IR)	0.000	3.062
LAVT-RIS(Input:VI)	0.787	2.820
VLT-RIS(Input:IR)	0.787	2.401
VLT-RIS(Input:VI)	0.787	1.910
MG-RIS(Input:IR)	25.984	27.811
MG-RIS(Input:VI)	33.070	34.777
Ours(Input:IR and VI)	73.228	66.065

Table 2. Time overhead comparison of various fusion models on two test datasets.

Test Sets	CDDFuse	DDcGAN	EMMA	MetaFusion	MUFusion
TNO	0.0319s	14.8790s	0.3181s	0.0040s	6.9263s
M ³ FD	0.0329s	26.5890s	0.4074s	0.0030s	19.2424s
Test Sets	MURF	SDNet	U2Fusion	FILM	Ours
TNO	0.0873s	0.0573s	4.3100s	0.6302s	0.3125s
M ³ FD	0.1701s	0.0584s	0.3583s	0.6384s	0.5587s

els (LAVT [9], VLT [2], MG [1]) on IR and VIS images. Unlike these models, which process only single images and yield suboptimal results due to modality limitations, our VIS-IR-tailored method takes two images as input to create a joint visual representation, reducing localization difficulty. Numerical results are listed in Table 1, demonstrating that our model is able to generate more accurate pseudo-labels. Qualitative comparisons are presented in Section 4.5 of the main paper.

7. Runtime Analysis

Table 2 lists the runtime of 10 fusion models. MetaFusion and CDDFuse are faster due to their lower computational complexity. Our model ranks 5th, with an acceptable runtime. The additional processing time is attributed to handling the extra text modality data, which is necessary for our model’s functionality. Experiments are conducted on a machine with an RTX 4090 GPU, i9-14900K CPU, 128GB RAM, and PyTorch 2.3.0.

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